

MURCIA SAN JAVIER AB, SpainWMO: **084330**Lat: **37.789N**Lon: **0.803W**Elev: **3**StdP: **101.28**Time Zone: **1.00 (EUC)**Period: **82-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	3.8	-4.7	2.5	9.1	-2.9	3.0	9.6	12.2	11.4	10.5	12.7	2.4	250	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.5	19.5	28.8	23.9	18.8	28.3	23.3	18.0	27.7	79.5	30.3	76.9	29.3	74.6	28.5	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	9.8	8.5	-0.7	35.7	1.7	1.6	-1.8	36.9	-2.8	37.8	-3.7	38.8	-4.9	39.9		
			-1.8	26.8	1.5	0.8	-2.9	27.4	-3.8	27.8	-4.7	28.3	-5.8	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	11.3	12.0	13.6	15.8	18.3	22.5	25.6	26.0	23.9	19.9	15.3	12.4	(6)
	DBStd	5.69	2.74	2.92	2.45	2.01	2.21	2.14	1.69	1.86	1.75	2.48	2.95	2.31	(7)
	HDD10.0	39	17	13	2	0	0	0	0	0	0	0	1	6	(8)
	HDD18.3	947	219	177	148	80	28	1	0	0	0	13	98	183	(9)
	CDD10.0	2989	56	69	114	173	259	375	482	495	417	308	160	81	(10)
	CDD18.3	856	0	1	1	3	28	126	224	237	167	63	7	0	(11)
	CDH23.3	6868	1	2	16	13	101	778	2139	2312	1200	275	30	1	(12)
	CDH26.7	1766	0	0	2	1	5	167	625	697	243	22	4	0	(13)
Wind	WSAvg	4.0	3.6	4.2	4.4	4.6	4.3	4.3	3.9	4.1	3.9	3.5	3.6	3.4	(14)
Precipitation	PrecAvg	280	33	22	26	27	21	8	1	5	37	31	36	35	(15)
	PrecMax	442	160	154	78	135	64	54	12	38	165	128	121	216	(16)
	PrecMin	117	0	0	0	2	0	0	0	0	3	0	4	0	(17)
	PrecStd	75	37	32	20	29	19	14	3	9	38	31	30	44	(18)
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.7	23.2	25.3	25.0	27.4	31.9	34.4	34.4	32.1	28.8	26.7	21.7	(19)
		MCWB	13.6	15.3	15.9	16.8	18.8	21.8	23.0	24.6	22.3	20.4	17.8	13.9	(20)
	2%	DB	19.7	20.3	22.7	23.1	25.8	29.9	32.1	32.3	30.2	27.0	23.6	19.6	(21)
		MCWB	13.2	13.3	15.3	15.5	18.3	21.7	23.4	24.5	22.7	20.8	17.2	13.9	(22)
	5%	DB	18.0	18.7	20.7	21.6	24.4	28.4	30.8	31.0	29.1	26.0	21.9	18.2	(23)
		MCWB	12.1	12.9	14.4	15.1	17.8	21.2	23.7	24.5	22.6	20.7	16.2	13.8	(24)
	10%	DB	16.5	17.1	19.0	20.2	23.1	27.2	29.8	30.0	28.0	25.0	20.3	17.1	(25)
		MCWB	11.7	12.4	13.4	14.6	17.7	20.8	23.9	24.2	22.6	20.1	16.1	13.5	(26)
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	16.0	17.7	18.2	20.6	24.3	26.2	26.8	25.3	23.0	19.9	16.5	(27)
		MCDB	19.3	20.5	22.8	21.9	25.2	29.1	30.4	31.4	29.4	26.1	24.3	18.2	(28)
	2%	WB	14.5	14.6	16.0	17.2	19.7	23.2	25.6	26.2	24.5	22.2	18.6	15.6	(29)
		MCDB	17.6	18.5	20.7	20.6	23.9	27.8	29.9	30.5	28.4	25.5	21.5	17.8	(30)
	5%	WB	13.4	13.8	15.1	16.4	19.1	22.5	25.0	25.5	23.9	21.6	17.8	14.9	(31)
		MCDB	16.3	17.3	19.6	19.8	22.9	26.7	29.1	29.6	27.6	24.9	20.7	17.4	(32)
	10%	WB	12.5	13.0	14.2	15.7	18.4	21.8	24.5	25.0	23.4	20.7	17.0	14.1	(33)
		MCDB	15.4	16.1	17.7	19.1	22.0	25.7	28.5	28.8	26.9	24.0	19.7	16.5	(34)
Mean Daily Temperature Range		MDBR	10.4	9.4	9.6	8.9	8.4	7.8	7.7	7.3	8.0	8.8	9.3	9.9	(35)
	5% DB	MCDBR	11.8	11.6	12.9	11.5	10.3	10.2	9.8	8.8	9.7	10.0	11.0	11.2	(36)
		MCWBR	7.1	7.1	7.7	6.6	5.8	5.4	5.4	4.9	5.2	5.7	6.7	7.1	(37)
	5% WB	MCDBR	10.5	10.4	11.5	9.4	8.4	8.3	8.1	7.8	8.2	8.9	8.6	10.0	(38)
		MCWBR	7.4	7.1	7.6	6.0	5.4	5.0	5.1	4.8	5.1	5.3	5.8	6.9	(39)
Clear-Sky Solar Irradiance	taub	0.317	0.342	0.389	0.401	0.428	0.450	0.463	0.463	0.431	0.396	0.351	0.324	(40)	
	taud	2.506	2.456	2.322	2.314	2.266	2.223	2.197	2.202	2.301	2.391	2.480	2.507	(41)	
	Ebn at Noon	869	887	871	879	857	835	820	811	817	813	819	833	(42)	
	Edh at Noon	79	94	118	126	135	141	144	140	120	100	81	75	(43)	
All-Sky Solar Radiation	RadAvg	2.56	3.40	4.60	5.70	6.72	7.52	7.40	6.47	5.07	3.74	2.66	2.22	(44)	
	RadStd	0.23	0.33	0.35	0.37	0.46	0.24	0.16	0.22	0.26	0.28	0.23	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	N/A	N/A	-1.30	+0.51	N/A	N/A	N/A	N/A	N/A	+58

Nomenclature: See separate page